MIL-W-52874B(ME)
7 March 1983
SUPERSEDING
MIL-W-52874A(ME)
25 May 1979

MILITARY SPECIFICATION

WATER QUALITY ANALYSIS SET: ENGINEER

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a water quality analysis set for use by engineer personnel.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-535	- Plastic Sheet (Sheeting); Plastic Strip; Vinyl Chloride Polymer and Vinyl Chloride - Vinyl Acetate Copolymer, Rigid.
NN-P-71	- Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way.
QQ-S-781	- Strapping, Steel, and Seals.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6630

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

MMM-A-1617	- Adhesive, Rubber Base, General Purpose.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-4150	- Cases, Transit and Storage, Water proof and Water-Vaporproof.
MIL-H-9884	- Honeycomb Material, Cushioning, Paper.
MIL-E-52798	- Enamel, Alkyd, Camouflage.
MIL-E-52929	- Enamel, Alkyd, Camouflage, Flash Dry.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-147	- Palletized and Containerized Unit Loads, 40 Inch X 48 Inch Pallets, Skids, Runners or Pallet Type Base.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-882	- System Safety Program for Systems and Associated Subsystems and Equipment: Requirements for.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

ME

TA 13222E7800	- Water Quality Analysis Set: Engineer

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:37Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN PUBLIC HEALTH ASSOCIATION (APHA)

Standard Methods for Examination of Water and Wastewater.

(Application for copies should be addressed to the American Public Health Association, 1015 18th St NW, Washington, DC 20036.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI Z129.1-1976 - Precautionary Labeling of Hazardous Industrial Chemicals.

(Application for copies should be addressed to American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The water quality analysis set (hereinafter called "set") shall consist of test kits, equipment, and reagent chemicals necessary to conduct water analysis in the field. The Water Quality Analysis Set, Engineer, shall be in accordance with Top Assembly TA 13222E7800 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the set. Any data (shop drawings, layout, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The contractor shall furnish one or more sets as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods and choice of design detail and components will produce sets that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

examination and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests, as specified (see 6.2).

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more sets for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.7).

3.4.1 Glass. Glass or other easily broken material shall not be used except for optical reasons or when specifically requested or approved by the Government.

3.4.2 Degradable materials. Materials that deteriorate after or are deformed by prolonged exposure to water or reagent chemicals supplied in the set shall not be used.

3.4.3 Chemical reagents. Chemical reagents may be either liquid or dry form. Liquid reagents used in the drop-count method of analysis shall be packaged in bottles with droppers or in self-contained dropper bottles. The reagents shall be formulated so that not more than 30 drops of liquid or 30 additions of dry reagents are required to determine any value within the range specified for each test.

3.4.3.1 Reagent. Potassium hydroxide reagent provided in the set shall be capped with an alkali resistant plastic screw cap. A separate dropper shall be provided for dispensing this reagent.

3.5 Test kits.

3.5.1 Turbidity test kit. The turbidity test kit (see drawing 13222E7799) shall consist of two plastic cylinders and the turbidity reagent. The method of test shall be a visual comparison of one turbidity cylinder containing the sample with the second turbidity cylinder containing turbidity-free water to which a known amount of turbidity reagent has been added. The kit shall provide for two turbidity ranges: One from 5 to 100 units in increments of 5 units and the other from 10 to 200 units in increments of 10 units. The test kit shall provide a capability to measure 0 to 200 Jackson turbidity units with standard turbidity reagent of a strength of 20 Jackson turbidity units per ml of reagent when used with a 25 ml test sample.

3.5.1.1 Performance. The test kit shall be provided with enough chemical reagents to perform not less than 100 tests at 10 units of turbidity.

3.5.2 Chloride test kit. The test kit (see drawing 13222E7778) shall provide the equipment and chemical reagents necessary to determine the concentration of sodium chloride (NACL) or chlorides (CL) in water in the specified range.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

3.5.2.1 High range. Concentration in the range of 0 - 2500 mg/l for sodium chloride or chlorides shall be determined using a silver nitrate solution drop count method with a chloride indicator. The test reagent shall be of such concentration that one drop shall be equal to 50 mg/l sodium chloride, or 30.3 mg/l chlorides, with a 5.83 ml sample.

3.5.2.2 Low range. Concentration in the range of 0 - 500 mg/l for sodium chloride or chlorides shall be determined using a silver nitrate solution drop count method with a chloride indicator. The test reagent shall be of such concentration that one drop shall be equal to 12.5 mg/l sodium chloride, or 7.58 mg/l chlorides, with a 23 ml sample.

3.5.2.3 Performance. The test kit shall be provided with enough chemical reagents to perform not less than 100 tests at 25 mg/l.

3.5.3 Sulfate test kit. The test kit (see drawing 13222E7796) shall provide the equipment and chemical reagents necessary to determine the concentration of sulfates in water in the range from 100 mg/l to 3000 mg/l. A drop count titration method with a sulfate indicator powder and standard barium chloride solution shall be used. The barium chloride solution shall be of such strength that 1 drop shall be equal to 25 mg/l sulfate when using a 10 ml sample.

3.5.3.1 Performance. The test kit shall be provided with enough chemical reagents to perform not less than 50 tests at 250 mg/l.

3.5.4 Hardness test kit. The test kit (see drawing 13222E7789) shall provide the equipment and chemical reagents necessary to determine the concentration of total hardness and calcium hardness in water in the range from 17 mg/l to 9000 mg/l. A drop count titration method will be used. The same titrating agent will be used for both calcium and total hardness determinations with the appropriate buffers and indicators. The indicator for the calcium determination shall be in the form of a powder pillow. The total hardness indicator used shall be provided as a solution of Erichrome Black T or equivalent. The titrating agent will be of such strength that 1 drop shall be equivalent to 17.2 mg/l as calcium carbonate when using a 5.83 ml test sample.

3.5.4.1 Performance. The test kit shall be provided with enough chemical reagents to perform not less than 100 tests at 250 mg/l of total hardness and 100 tests at 250 mg/l calcium hardness. 1/

3.5.5 Alkalinity test kit. The test kit shall provide the equipment and chemical reagents necessary to determine the total (T) and phenolphthalein (P) alkalinity of water in the range from 5 mg/l to 500 mg/l (as $CaCO_3$). A titration method employing a microburette shall be used with phenolphthalein indicator for (P) alkalinity and mixed bromcresol green/methyl red indicator for the total (T) alkalinity. A standard sulfuric acid solution shall be used as the

1/ The magnesium hardness is calculated by subtracting the calcium hardness from the total hardness.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

titrating agent. The strength of the titrating agent shall be such that 20 drops of reagent (using a microburette) shall equal a "P" alkalinity of 8 mg/l as calcium carbonate when using a 15 ml sample.

3.5.5.1 Performance. The test kit shall be provided with enough chemical reagents to perform not less than 100 tests at 120 mg/l T and P.

3.5.6 Color test kit. The test kit (see drawing 13222E7797) shall consist of a color comparator, APHA platinum-cobalt color disk with accessory components and shall have the capability of measuring color in the ranges from 0 to 100 units and 0 to 500 units.

3.5.6.1 Performance. The test kit shall be provided with sufficient reagent (color disc) to perform not less than 100 tests at 500 units.

3.5.7 Conductivity test kit. The conductivity (total dissolved solids) test kit (see drawing 13222E5258) shall consist of a compact meter, operating on an electrical conductivity principle, reading directly in parts per million from 1 to 5000 on three ranges, 0 - 50, 0 - 500, and 0 - 5000. And, with a ranger extension accessory, increases its range by a factor of 10. Power source shall be an internal mounted 9 volt transistor type battery with sufficient life to meet the performance criteria. Also, included, will be an instruction card per drawing 13222E7805.

3.5.7.1 Performance. The test kit shall have the capacity to perform 1000 water sample tests prior to replacement of the battery.

3.5.8 Flocculant test kit. The test kit shall provide the equipment and chemical reagents necessary for determination of chemical dosage requirements and coagulation test. The set's components which make up this test kit shall consist of and be in accordance with Jar Assembly (13222E7786), Measuring Spoon (13222E7793), Chlorine Demand Tablets (13222E7779), Ferric Chloride (13222E7784) and Powdered Limestone (13222E7791) and used with the Color Comparator. Instructions card in accordance with 13222E7803 shall be supplied.

3.5.8.1 Performance. The test kit shall be provided with sufficient chemical reagents and strength to conduct a minimum of 100 chlorine demand tests and for 100 coagulation tests.

3.6 Carrying case.

3.6.1 Set case. All items comprising a set shall be packaged in a case conforming to MIL-C-4150, type 1, style 1, class A, variety NHC except those changes as specified in the drawings or as specified herein. Cushioning material as specified in the drawings, molded or cut-to-form, shall be used to provide shock and vibration protection to the contents of the case during transport. All items contained in the case shall be easily removed and replaced without damage to any item or the cushioning material.

3.6.1.1 Set case finish. Unless otherwise specified (see 6.2), the case shall be finished in forest green in accordance with MIL-T-704. Metallic cases

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

shall use the type D finishing system from table II of MIL-T-704 except that the finish shall be in accordance with either MIL-E-52798 or MIL-E-52929. Glass fiber reinforced plastics or other compounded non-metallic materials used for cases shall use the type C finishing system from table II of MIL-T-704 except that the final finish shall be in accordance with either MIL-E-52798 or MIL-E-52929. Glass fiber reinforced plastics or other compounded non-metallic materials used for cases shall use the type C finishing system from table II of MIL-T-704 except that the final finish shall be in accordance with either MIL-E-52798 or MIL-E-52929. Non-metallic cases having the color pigmentation compounded in the material shall not require painting. A sample of either the forest green paint or of the case material having the color pigmentation compounded in it shall be submitted to the following laboratory for approval prior to acceptance of the initial production set: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-VO, Fort Belvoir, VA 22060. A minimum of a quart sample of paint from each production lot used must be forwarded. A sample of case material having the color pigmentation compounded in it shall be at least 2 inches square.

3.6.1.2 Catch assemblies. Unless otherwise specified (see 6.2), the catch assemblies per drawing 13222E7777 after treatment shall be painted at the option of the contractor.

3.6.2 Test-kit case. The components of the test kits for turbidity, low-range chloride, sulfate, hardness, alkalinity, and color shall be contained in impact-resistant plastic cases that are capable of being hand carried. Each case shall contain all the equipment required for conducting the test plus enough chemical reagents to conduct not less than 20 tests. Additional quantities of chemical reagents required to complete 100 tests shall be stored in the set case. The kit cases shall be equipped with a molded or cut-to-form cushioning insert to provide shock and vibration protection and to permit easy removal and replacement of the components without damage to any component or to the cushioning material. The case shall be provided with latches that will withstand repeated use and will not be prone to accidental opening.

3.7 Environmental.

3.7.1 Operating climatic conditions. The sets shall be capable of operating under the climatic conditions specified in table I when tested in accordance with 4.7.2.2.1.

TABLE I. Operating climatic conditions.

Climatic category	Ambient air temp° F	Solar radiation Btu/ft^2/hr	Ambient relative humidity, percent
1	Nearly constant 75	Negligible	95 to 100
2	78 to 95	0 to 360	74 to 100
5	70 to 110	0 to 360	20 to 85

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:37Z
Check the source to verify that this is the current version before use.

3.7.2 Storage climatic conditions. The sets shall be capable of storage under the climatic conditions specified in table II when tested in accordance with 4.7.2.2.2.

TABLE II. Storage climatic conditions.

Climatic category	Induced air temp ° F	Induced relative Humidity, percent
1	Nearly constant 80° F	95 to 100
2	90 to 125° F	10 to 85
5	70 to 145	5 to 50
6	-10 to -30	Tending toward saturation

3.8 Marking.

3.8.1 Set cases. Each set shall be identified in accordance with MIL-STD-130. The identification marking shall be applied to the set case as specified on the drawings. The plates shall be attached to the case so that there will be no ingress of moisture and located so as to be both visible and legible.

3.8.1.1 Stenciling. Case tops will be stenciled with standard white stencil ink with the following "Storage Temp Maximum 125° F". Stenciling will be applied on the case top siding above the pressure relief valve with 1 inch high block letters.

3.8.2 Test-kit cases. Each kit case shall be identified in accordance with MIL-STD-130. The identification marking shall be applied to the case as specified herein and on the drawings. The identification shall be visible and legible.

3.8.3 Reagent chemicals. Each container of reagent chemicals shall be labeled with the generic name or manufacturer's trade name, use, and date of manufacture. Any reagent chemicals which according to the manufacturer have a limited shelf life shall have printed on the label the expiration date of the reagent.

3.8.3.1 Cautions and warnings. Each container of reagent chemicals shall be labeled with cautions and warnings where contact with, inhalation of, or ingestion of the reagent would have an adverse effect on the user. Such information shall include but not be limited to the following:

(a) Generic name of the reagent
(b) Symptoms
(c) Antidote
(d) Signal words (i.e., danger, warning, caution, etc.)
(e) Statement of hazards (i.e., flammable, harmful, etc.)

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:37Z
Check the source to verify that this is the current version before use.

Labeling of chemicals shall be in accordance with the requirements given in American National Standards Institute Bulletin ANSI Z129.1-1976, Precautionary Labeling of Hazardous Industrial Chemicals.

3.9 Instructions.

3.9.1 Loading diagram. The set case shall be provided with a loading diagram depicting the proper location and quantities of each item in the case. The diagram shall be as specified on the drawing 13222E7802.

3.9.2 Instructions for test kits. Each kit shall be provided with instructions that contain detailed procedures for conducting the test. These instructions shall be printed on material which is water resistant or material which has been water resistant coated. Markings shall be in accordance with MIL-STD-130 and as specified on the drawings. These instructions shall be affixed to the interior lid of the kit with an adhesive conforming to MMM-A-1617. When this is not practical, the instructions shall be printed on material conforming to L-P-535, composition B, type I, class 3, matte finish, 0.020 inch thick (minimum) black printing on opaque white, on a sheet of material 8 inches wide x 10-1/2 inches long.

3.9.3 Instructions for determination of chemical dosage requirements. Instructions for determining chemical dosage requirements shall be as specified in the drawings. These instructions shall be stored in the set case.

3.9.4 Instruction for turbidity-free water. Instructions for preparing turbidity-free water shall be as specified in the drawings. These instructions shall be stored in the set case.

3.10 Government-furnished property. When specified (see 6.2), the following property in the quantities indicated will be furnished by the Government for overpack in the set case (see 6.4):

Item No.	Description	Identification	Quantity for each set
1.	Comparator, Assembly	NSN 6630-01-044-0334	1
2.	Chemical Agent, Water Test (AN-M2)	NSN 6665-00-171-9747	1
3.	Refill Kit, Chemical Agent, Detection (ABC-M30)	NSN 6665-00-909-3647	1
4.	Tablets DPD No. 1	NSN 6810-01-044-0315	100 ea.
5.	Job Aids	N/A	1

3.11 Transportability. The set shall be capable of withstanding the vibrational stress and shock encountered when transported as loose cargo and airdrop delivery.

3.12 Weight and configuration. The set case and contents shall not exceed 65 pounds and shall be configured so that it can be easily carried by one man.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

3.13 Safety. The components of the system must not present any hazards to personnel during production or use per MIL-STD-882.

3.14 Human factors. The set shall conform to the human factors engineering requirements of MIL-STD-1472 and 3.4.3, 3.6, 3.8, 3.9, and 3.12 of this specification.

3.15 Workmanship. The cases and all component parts shall be fabricated free from any defects which could affect reliability, strength, or serviceability. The cushioning material installed in the cases shall be uniformly molded, formed, or cut. All markings, labels, and printed matter shall be legible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.1.2 End item final inspection report (EIFIR). Unless otherwise specified (see 6.2), each water testing kit shall be inspected in accordance with the applicable EIFIR (EF13222E7800). One copy of the completed EIFIR shall be shipped with each inspected Water Quality Analysis Set.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.6).
(e) Inspection of packaging (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction set shall be examined for the defects marked "X" in column 1 of table III. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The preproduction set shall be subjected to the tests marked "X" in column 1 of table IV. Failure of any test shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production sets will be selected at random by the Government from sets being produced by production tooling and will be examined for the defects marked "X" in column 2 of table III and subjected to the tests marked "X" in column 2 of table IV to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production set shall not exclude the remaining sets from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any or all other tests contained in this specification as part of the initial production test, and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production set to meet any requirements specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production set and shall be cause for refusal by the Government to continue acceptance of production sets until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on sets previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all sets accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.5.2 Examination. Samples selected in accordance with 4.5.1 shall be examined for the major and minor defects marked "X" in column 3 of table III. AQL shall be 2.5 percent defective for major defects and 6.5 percent defective for minor defects.

4.5.3 Tests. Samples selected in accordance with 4.5.1 shall be subjected to the tests marked "X" in column 3 of table IV. AQL shall be 2.5 percent defective.

4.6 Inspection comparison. The Government may select sets at any time during the contract production period and subject these sets to the examination marked "X" in column 4 of table III and the tests marked "X" in column 4 of table IV to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on sets selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production sets. In addition to any test specified as part of the inspection

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:37Z
Check the source to verify that this is the current version before use.

comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.6.1 Inspection failure. Failure of an inspection comparison set to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison set and shall be cause for refusal by the Government to continue acceptance of production sets until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on sets previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all sets accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.7 Inspection schedule.

4.7.1 Examination.

TABLE III. Examination schedule.

Preproduction	Initial Production	Quality Conformance	Comparison	Defect	Requirement Paragraph
1	2	3	4	5	6
				Major	
X	X	X	X	101. Clasps not as specified.	3.6.1
X	X	X	X	102. Cushioning insert malformed or not cut to fit.	3.6.2
X	X	X	X	103. Latches not as specified.	3.6.2
X	X	X	X	104. Components missing or not as specified.	3.5
X	X	X	X	105. Materials not as specified.	3.4
X	X	X	X	106. Components broken or damaged so as to affect serviceability.	3.4.1
X	X	X	X	107. Materials showing evidence of degradation.	3.4.2

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

TABLE III. Examination schedule. (Cont'd)

Preproduction	Initial Production	Quality Conformance	Comparison	Defect	Requirement Paragraph
1	2	3	4	5	6
X	X	X	X	108. Cautions, warnings, or antidote information missing, illegible, or incomplete.	3.8.3.1
X	X	X	X	109. Procedural instructions missing or inadequate.	3.9
X	X	X	X	110. Weight limits exceeded.	3.12
X	X	X	X	111. Improper configuration for one-man carry.	3.12
X	X	X	X	112. Workmanship not as specified.	3.15
X	X	X	X	113. Expiration date of reagent exceeded.	3.8.3
X	X	X	X	114. Case not finished as specified.	3.6.1.1
X	X	X	X	115. Labels missing, not as specified, without complete information or not legible.	3.8.3
				Minor	
X	X	X	X	201. Identification plates missing, not as specified, not attached as specified, attached in obscure location, or not legible.	3.8
X	X	X	X	202. Loading diagram not as specified, missing, incomplete, or not legible.	3.9.1

4.7.2 Tests.

4.7.2.1 Test schedule.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

TABLE IV. Test schedule.

Preproduction	Initial Production	Quality Conformance	Comparison	Tests	Test Paragraph	Requirement Paragraph
1	2	3	4	5	6	7
X	X	X	X	Environmental	4.7.2.2	3.7
X	X	X	X	Transportability	4.7.2.3	3.11
X	X	X	X	Operational (accuracy)	4.7.2.4	3.5.1 thru 3.5.8
X	X	X	X	Safety	4.7.2.5	3.13
X	X	X	X	Human factors	4.7.2.6	3.14

4.7.2.2 Environmental.

4.7.2.2.1 Operation temperature. A certificate stating that test kits shall be operational in the climatic conditions stated in table I will be acceptable in lieu of testing.

4.7.2.2.2 Storage. Each sample set shall be tested in accordance with MIL-STD-810, method 501, procedure II and method 502, procedure I, with the following exceptions:

Method 501 Procedure II

Step 4 - Change 68° C (154° F) to 52° C (125° F).
Step 6 - Repeat steps 3, 4, and 5 one additional time (making a total of two 12 hour cycles).
Step 7 - Delete.
Step 8 - Delete.

Method 502 Procedure I

Step 2 - Lower the internal chamber temperature to -35° C (-30° F) for six continuous hours with no wind or solar radiation, and with humidity tending towards saturation.
Step 3 - Delete.
Step 4 - Raise the internal chamber temperature to -23° C (-10° F) and maintain for 6 hours.
Step 5 - Repeat steps 2 and 4 one additional time (making a total of two 12 hour cycles).

The test sets will be transferred directly from the conditions of step 11, procedure II, method 501 to step 1, procedure I, method 502. Upon completion of this test each sample set will be inspected for damage to case and contents. Any evidence of permanent deformation, cracks, or broken items shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

4.7.2.3 Transportability.

4.7.2.3.1 Loose cargo. Each sample set shall be tested in accordance with MIL-STD-810, category g, method 514.1, procedure XI (loose cargo) and method 516.1 procedure II (transient drop test). Upon completion of this test each sample set shall be examined for damage to the case and contents. Any evidence of permanent deformation, cracks, or broken items shall constitute failure of this test.

4.7.2.3.2 Airdrop. When specified (see 6.2), each sample set shall be tested for airdrop capability and shall be prepared for test by strapping the test set with all components in place to a wood platform (approximately 30 inches x 30 inches) with energy dissipator material (MIL-H-9884, class 3) between the sample set and the wood platform. The total surface area of the dissipator material in contact with the test set shall be 64 square inches consisting of 8 pieces of dissipator material 4 inches x 4 inches x 3 inches high. The dissipator material shall be arranged in four stacks of material each two pieces high. The stacks shall be glued to each other and to the case on the bottom edges of the case midway between the corners (fig. 1). The strapped case with all components, dissipator material, and wood platform shall be free dropped from a height of 12 feet 8 inches (152 inches) onto a flat rigid concrete surface. The drop shall be oriented so that the platform with test assembly shall fall flat at impact. Upon test completion, the set shall be examined for damage to case and contents. Any evidence of permanent deformation, cracks, or broken items shall constitute failure of this test.

4.7.2.4 Operational (accuracy). When specified (see 6.2), the following kits shall be removed from the sample sets and tested for the requirements as stated in the applicable paragraph as shown below. Unless otherwise specified (see 6.2), the contractor shall request from each vendor/supplier of chemicals, copies of their test data on each batch/distinct lot furnished to confirm that the chemicals meet specification requirements.

(a) Turbidity test kit (3.5.1)
(b) Chloride test kit (3.5.2)
(c) Sulfate test kit (3.5.3)
(d) Hardness test kit (3.5.4)
(e) Alkalinity test kit (3.5.5)
(f) Color test kit (3.5.6)
(g) Conductivity test kit (3.5.7)
(h) Flocculant test kit (3.5.8)

Test solutions will be prepared at each of the decision concentrations associated with each individual kit (i.e., 250, 400 and 2,700 mg/l in the case of sulfates.) Replicate measurements will be conducted and the resulting precision will be calculated. Failure to meet applicable precision criteria shall constitute failure of this test.

4.7.2.5 Safety. The set shall be evaluated throughout all testing to determine compliance with 3.13. Nonconformance to 3.13 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:37Z
Check the source to verify that this is the current version before use.

4.7.2.6 Human factors. The set shall be evaluated throughout all testing to determine compliance with 3.14. Nonconformance to 3.14 shall constitute failure of this test.

4.8 Inspection of packaging.

4.8.1 Quality conformance inspection of pack.

4.8.1.1 Unit of product. For purposes of inspection, the unit of product shall be a completed pack of a single set prepared for shipment.

4.8.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.1.3 Examination. Samples selected in accordance with 4.8.1.2 shall be examined for the following major defects. AQL shall be 2.5 percent defective. All completed pallet loads shall be examined for the applicable defects.

116. Set components not fitted securely in designated spaces within carrying case.
117. Carrying-case clasps not properly latched.
118. Shipping container not as specified for levels A and B.
119. Shipping container not strapped as specified for levels A and B.
120. Palletized loads not on type of pallet specified for levels A, B, or commercial.
121. Sets not secured to pallets as specified for all degrees.
122. Marking illegible, incomplete, incorrect, or missing.

5. PACKAGING

5.1 Preservation, level A, level B and commercial. The carrying case shall be checked to insure that all set components are securely fitted into their designated locations. All clasps shall be securely latched.

5.2 Packing. Packing shall be level A, level B or commercial as specified (see 6.2).

5.2.1 Level A. Each carrying case containing a complete set shall be packed in a snug-fitting box conforming to PPP-B-601, overseas type, style I. The box shall be reinforced in accordance with the box specification with flat steel strapping conforming to QQ-S-781, class 1, type I or IV, minimum size not less than 1/2 inch wide x 0.02 inch thick; and, unless otherwise specified herein, shall be finish B. When specified (see 6.2), strapping shall be finish A.

5.2.2 Level B. Each carrying case containing a complete set shall be packed in a box conforming to PPP-B-636, type CF, class weather resistant, grade V3c. The box shall be closed, sealed, and reinforced in accordance with the box specification.

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:37Z
Check the source to verify that this is the currentversion before use.

5.2.3 Commercial. The carrying case containing a complete set shall be packed in accordance with MIL-STD-1188. The carrying case shall have tamperproof seals attached to two clasps, one each on opposite faces of the carrying case.

5.3 Palletization. When four or more sets are to be shipped to the same destination, packed carrying cases shall be palletized. For levels A and B, pallets shall conform to NN-P-71, type IV. Loading and securing of carrying cases shall be in accordance with MIL-STD-147. For commercial, pallets shall be a commercial expendable wood type, 40 inches x 48 inches in size. Flat steel strapping shall be used to secure carrying cases to the pallet.

5.4 Marking. Marking for level A and level B shall be in accordance with MIL-STD-129. Commercial marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The water quality analysis set is intended for use by engineer personnel to provide immediate information concerning the presence of chemical contamination in order to determine the type of water purification equipment required and to maintain and control optimum operating conditions of this equipment.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
(c) Time frame required for submission of preproduction model, and number of sets required (see 3.2).
(d) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
(e) When initial production inspection is required and number of sets to be furnished (see 3.3).
(f) When color and finish is other than as specified (see 3.6.1.1).
(g) When painting of catch assemblies is required (see 3.6.1.2).
(h) When Government-furnished property is required (see 3.10 and 6.4).
(i) When an EIFIR is not required for each set (see 4.1.2).
(j) When set is to be air drop tested (see 4.7.2.3.2).
(k) When kit shall be removed and tested (see 4.7.2.4).
(l) When vendor data is not required (see 4.7.2.4).
(m) Degree of packing required (see 5.2).
(n) When strapping shall be finish A (see 5.2.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

6.3 Preproduction model. Any changes or deviations of production sets from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish sets conforming to this specification (see 3.2).

6.4 Government-furnished property. The contracting officer should arrange to furnish the property specified in 3.10, when applicable (see 6.2).

6.5 Supplies and equipment. The contracting officer should furnish the contractor the list of supplies and equipment as specified in SC 6630-97-CL-E07, NSN 6630-00-140-7820.

6.6 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, and contractors operation manual to be furnished with each set.

6.7 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.4).

Custodian:
Army - ME

Preparing activity:
Army - ME

Review activity:
Army - MD

Project 6630-A325

☆ U.S. GOVERNMENT PRINTING OFFICE: 1983—605-034/1679

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

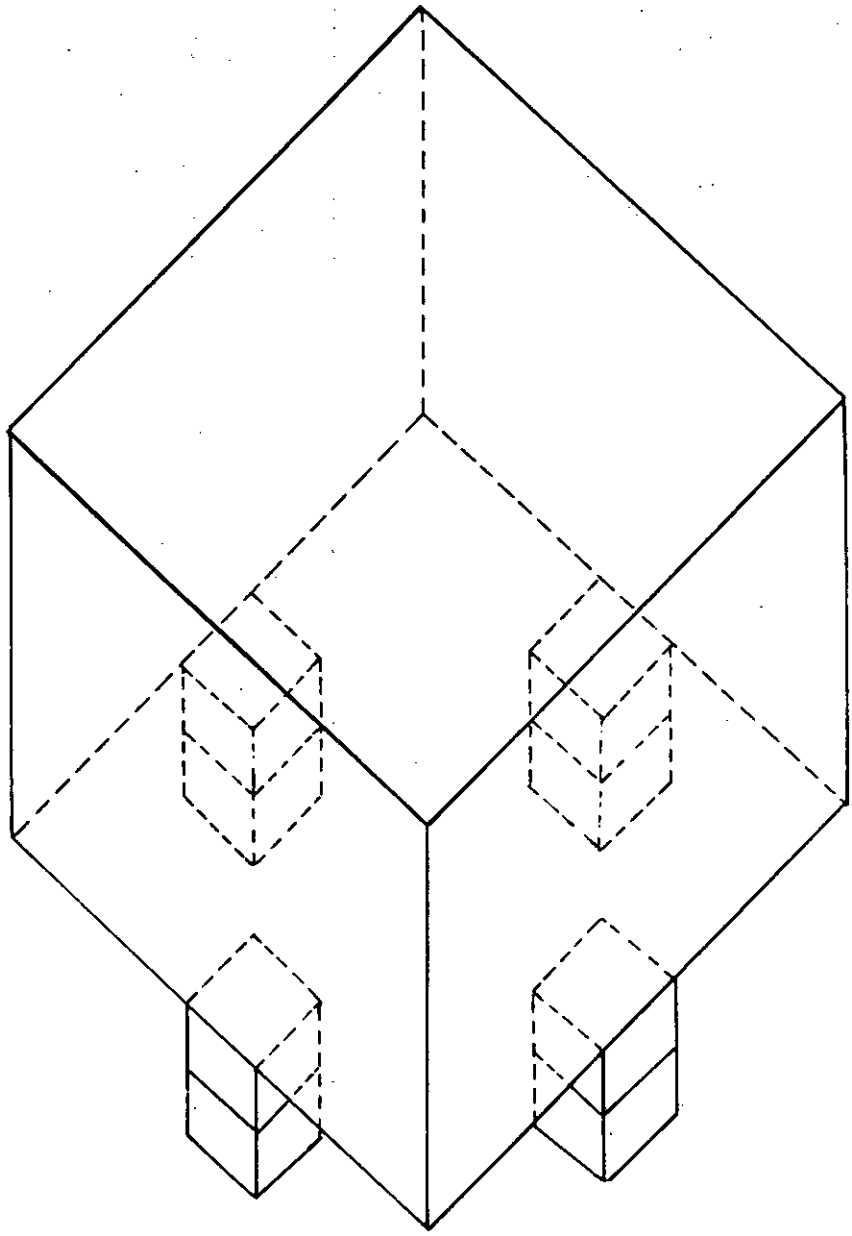


FIGURE 1. Energy dissipator material.

X-3527

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY CARD

FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

COMMANDER
US ARMY MOBILITY EQUIPMENT RESEARCH
AND DEVELOPMENT COMMAND
ATTN: DRDME-DS
FORT BELVOIR, VA 22060

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:37Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-W-52874B(ME) Water Quality Analysis Set: Engineer

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:37Z
Check the source to verify that this is the current version before use.